

Simulating Spectral Sampling of the MICA Files with the Enfys IR Spectrometer for the ExoMars Rosalind Franklin Rover

March 14, 2025

1 Abstract

Here we investigate the expected ability of a new remote-sensing stand-off infrared spectrometer, *Enfys*, for the ExoMars Rosalind Franklin Rover, to resolve diagnostic features of the reflectance spectra of key minerals previously identified on the surface of Mars.

2 Introduction

2.1 Background

ESA plans to land the ExoMars Rosalind Franklin Rover at the Oxia Planum region of Mars¹ to characterise the geochemical and aqueous history of the environment and near subsurface (<2 m), and to search for signs of past or present life².

Previously the rover payload included a ROSCOSMOS (Russian) led instrument, the Infrared Spectrometer for ExoMars (ISEM)³. Following the geopolitical fallout of the Russian invasion of Ukraine in 2021 the ISEM instrument was removed from the rover. A new UK-led instrument, *Enfys* (Welsh for ‘rainbow’), has been approved at the ESA ministerial level to replace it.

The purpose of the spectrometer is to provide high spectral resolution observations of points of interest found in panoramic colour and multispectral image products, captured by the mast-mounted PanCam instrument⁴ of the rover. The short-wave infrared range is sensitive to the IR-active H₂O, OH, and CO₃ molecules found in the carbonates and hydrated clays, sulphates and silicates that

¹Quantin-Nataf, C., Carter, J., Mandon, L., Thollot, P., Balme, M., Volat, M., Pan, L., Loizeau, D., Millot, C., Breton, S., Dehouck, E., Fawdon, P., Gupta, S., Davis, J., Grindrod, P. M., Pacifici, A., Bultel, B., Allemand, P., Ody, A., ... Broyer, J. (2021). Oxia Planum: The Landing Site for the ExoMars “Rosalind Franklin” Rover Mission: Geological Context and Prelanding Interpretation. *Astrobiology*, 21(3). <https://doi.org/10.1089/ast.2019.2191>

²Vago, J. L., Westall, F., Pasteur Instrument Teams, Landing S, Coates, A. J., Jaumann, R., Korablev, O., Ciarletti, V., Mitrofanov, I., Josset, J.-L., De Sanctis, M. C., Bibring, J.-P., Rull, F., Goesmann, F., Steininger, H., Goetz, W., Brinckerhoff, W., Szopa, C., Raulin, F., Westall, F., ... the ExoMars Project Team. (2017). Habitability on Early Mars and the Search for Biosignatures with the ExoMars Rover. *Astrobiology*, 17(6–7), 471–510. <https://doi.org/10.1089/ast.2016.1533>

³Korablev, O. I., Dobrolensky, Y., Evdokimova, N., Fedorova, A. A., Kuzmin, R. O., Mantsevich, S. N., Cloutis, E. A., Carter, J., Poulet, F., Flahaut, J., Griffiths, A., Gunn, M., Schmitz, N., Martín-Torres, J., Zorzano, M.-P., Rodionov, D. S., Vago, J. L., Stepanov, A. V., Titov, A. Yu., ... Ivanov, A. Yu. (2017). Infrared Spectrometer for ExoMars: A Mast-Mounted Instrument for the Rover. *Astrobiology*, 17(6–7), 542–564. <https://doi.org/10.1089/ast.2016.1543>

⁴Coates, A. J., Jaumann, R., Griffiths, A. D., Leff, C. E., Schmitz, N., Josset, J.-L., Paar, G., Gunn, M., Hauber, E., Cousins, C. R., Cross, R. E., Grindrod, P., Bridges, J. c., Balme, M., Gupta, S., Crawford, I. A., Irwin, P., Stabbins, R., Tirsch, D., ... the PanCam Team. (2017). The PanCam Instrument for the ExoMars Rover. *Astrobiology*, 17(6–7), 511–541. <https://doi.org/10.1089/ast.2016.1548>

have been found previously on Mars, the presence of which can give insights to the geochemical and aqueous history of the environment. The high spectral resolution is required to resolve the subtle changes in the IR-active molecule band depths, positions and asymmetries that are induced by the presence of the various other elements and molecules in different mineral groups, species and phases.

2.2 Enfys

The *Enfys* infrared spectrometer uses a scanning linear-variable-filter (LVF) optical design to observe incident narrowband intensity in the spectral range of 0.9 - 3.1 μm with a spectral resolving power of ~ 100 , and with an expected signal-to-noise ratio of >2000 at 0.9 μm and >100 at 3.1 μm . The spectral range is covered by two photodiodes and two separate linear-variable filters; a 0.9 - 1.7 μm range filter is paired with an uncooled InGaAs photodiode, and the 1.6 - 3.1 μm range filter is paired with a cooled TE photodiode. The internal design of the system is shown in figure 1.

When an LVF of spectral range $\lambda_{lo} - \lambda_{hi}$ of active length L is positioned such that the instrument optical path intersects the point $x \in [0, L]$, a transmission profile with centre-wavelength $\lambda_{cwl} \in [\lambda_{lo}, \lambda_{hi}]$ attenuates the transmitted spectral radiance. The ‘linear’ part of the LVF name refers to the linear relationship between Δx and $\Delta \lambda_{cwl}$. Typically, we expect the full-width-at-half-maximum to follow a linear relationship with $\lambda_{cwl} \forall \lambda_{cwl} \in [\lambda_{lo}, \lambda_{hi}]$, given by the spectral resolving power $\lambda/\Delta\lambda$. $\lambda/\Delta\lambda$ is a property of the filter and of the diameter of the optical beam path at the point of intersection (a larger beam path gives a lower spectral resolving power).

```
[1]: from IPython.display import display, Image, HTML
display(Image('enfys_diagram.png'))
```

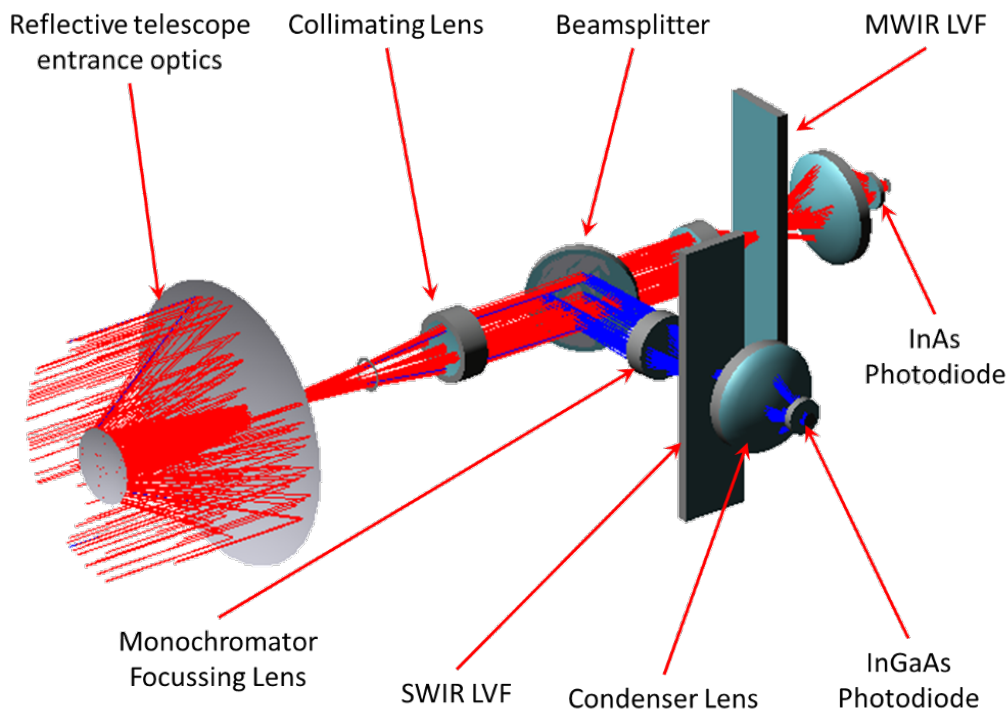


Figure 1 Schematic of the *Enfys* IR spectrometer optical path, showing the two Linear Variable Filters (SWIR LVF, MWIR LVF), and the two separate photodiodes.

2.3 Problem

This optical design is functionally different to the Acousto-Optic Tunable Filter design of ISEM, so it is important that the design is demonstrated to fulfil the performance requirements of the instrument, and that expectations of performance can be set prior to laboratory tests of the completed instrument.

At time of writing, the spectrometer is in the development stage. This notebook describes how the Spectral Parameters Toolkit⁵ can be used to convert the developing expectations of the spectral range and resolution and radiometric sensitivity of the instrument into predictions of its ability to discriminate mineral species and phases on the Mars surface at the rover landing site.

3 Method

To investigate the expected performance we use high-resolution laboratory measured spectra of a range of minerals expected at the Mars surface, and computationally simulate the sampling of the spectra with the expected spectral response of *Enfys*. We will perform repeat sampling with synthetically added noise to investigate the spectral and radiometric limits of the instrument.

We will produce a number of visualisations to investigate the predicted instrument performance, and demonstrate the limits of the instrument discriminatory ability.

3.1 Spectral Resampling

We take a simple approach to resampling, as follows.

We neglect reflectance calibration uncertainty, and any systematic uncertainty when merging the data from the two separate photodiodes, and assume that the linear-variable filter movement is perfectly calibrated such that a measurement at centre-wavelength λ_{cwl} can be requested with arbitrary precision.

We assume an observation of λ_{cwl} has a Cauchy transmission profile, and, prior to component characterisation, a spectral resolving power of $\lambda/\Delta\lambda = 100$ is achieved for all λ_{cwl} , such that the Cauchy transmission profile for λ_{cwl} has a full-width-at-half-maximum of $\Delta\lambda = \lambda_{cwl}/100$.

The Cauchy transmission profile $T(\lambda|\lambda_{cwl})$ is given by

$$T(\lambda|\lambda_{cwl}) = \frac{\gamma^2}{(\lambda - \lambda_{cwl})^2 + \gamma^2}$$

where $\gamma = \Delta\lambda/2 = (\lambda_{cwl}/100)/2$.

We assume that a high-resolution laboratory spectrum approximates the continuous function $R(\lambda)$, such that the calibrated reflectance at λ_{cwl} has the expected value $R[\lambda_{cwl}]$ of:

⁵Stabbins, R., & Grindrod, P. (2024). The Spectral Parameters Toolkit (sptk): Release v0.1 (Version v0.1) [Computer software]. Zenodo. <https://doi.org/10.5281/zenodo.10694286>

$$R[\lambda_{cwl}] = \frac{\int R(\lambda)T(\lambda|\lambda_{cwl})d\lambda}{\int T(\lambda|\lambda_{cwl})d\lambda}$$

To simulate noise we add ε , an additive noise term that we draw from the Gaussian distribution with variance:

$$\sigma^2 = \frac{\sqrt{R[\lambda_{cwl}]}}{\text{SNR}}$$

under the assumption that maximum SNR is only achieved for $R = 1$, and the noise is dominated by counting noise, such that σ follows a Poissonian relationship with the signal R .

For each observation $R[\lambda_{cwl}]$ we draw 100 repeat samples, such that

$$R_i[\lambda_{cwl}] = R[\lambda_{cwl}] + \varepsilon_i$$

with

$$\varepsilon_i \sim \mathcal{N}\left(0, \frac{\sqrt{R[\lambda_{cwl}]}}{\text{SNR}}\right)$$

This gives an indication of the expected noise of a calibrated *Enfys* acquired spectrum.

3.2 Analysis

For now, we simply produce plots of the resampled spectra, and visually inspect the plots to consider the ability to resolve diagnostic spectral features against the noise. We will include more details of analysis as that is developed here.

3.3 Notebook and Environment Setup

Here we include the required code blocks for setting up this notebook for this study.

```
[2]: %load_ext autoreload
      %autoreload 2
      %config InlineBackend.figure_format = 'svg' # note - requires Inkscape to be
      ↪ installed, and to be accessible from the terminal
      # on macos, add inkscape to the path with `sudo ln -s /Applications/Inkscape.
      ↪ app/Contents/MacOS/inkscape /usr/local/bin/inkscape`
      %matplotlib inline
```

We will be using the Spectral Parameters Toolkit. First we import the required modules.

```
[3]: import sptk.config as cfg
      from sptk.material_collection import MaterialCollection
      from sptk.instrument import Instrument
      from sptk.observation import Observation
      from sptk.spectral_library_analyser import SpectralLibraryAnalyser as sla
```


We set our simulation resolution range to extend the range of Enfys, of 0.9 - 3.1 μm , such that the long tails of the Cauchy distribution of the linear-variable-filter profiles can be accommodated during sampling. We leave our spectral resolution at 1 nm for all λ .

```
[4]: cfg.update_sample_res('wvl_min', 850)
      cfg.update_sample_res('wvl_max', 3300)
```

Finally we set the project name.

```
[5]: PROJECT_NAME = 'enfys_mica'
```

4 Data

For this study we are using a spectral library of minerals identified through CRISM analysis ([the MICA files](#))⁶. Here we use the laboratory spectra that have been identified that best represent the type localities presented in the MICA files. These can be accessed [here](#), and are sourced from the RELAB and USGS public spectral libraries.

We have grouped the spectral library into categories as follows, and included this categorisation in the `sptk` distribution, accessed from the `sptk.material_sets` module.

```
[6]: from sptk.material_sets import MICA_SET
```

We use the `sptk.material_collection` module to parse the spectral library:

```
[7]: matcol = MaterialCollection(
      MICA_SET,
      'MICA_new_dirtree',
      PROJECT_NAME,
      load_existing=False,
      balance_classes=False,
      random_bias_seed=None,
      allow_out_of_bounds=True,
      plot_profiles=False,
      export_df=True)
```

Building new enfys_mica MaterialCollection DF

Category

species|subgroup|group|
data_id status

Iron Oxides & Primary Silicates

-clinopyroxene|pyroxenes|inosilicates|
-NBPP22.csv loaded

⁶Viviano-Beck, C. E., Seelos, F. P., Murchie, S. L., Kahn, E. G., Seelos, K. D., Taylor, H. W., Taylor, K., Ehlmann, B. L., Wiseman, S. M., Mustard, J. F., & Morgan, M. F. (2014). Revised CRISM spectral parameters and summary products based on the currently detected mineral diversity on Mars. *Journal of Geophysical Research: Planets*, 119(6), 1403–1431. <https://doi.org/10.1002/2014JE004627>

-orthopyroxene|pyroxenes|inosilicates|
-LAPP47B.csv loaded
-fayalite|olivines|nesosilicates|
-LAP005.csv loaded
-forsterite|olivines|nesosilicates|
-LASC02.csv loaded
-hematite|hematites|oxides|
-GDS27.csv loaded
-plagioclase|feldspars|tectosilicates|
-NCLS04.csv loaded

Sulfates

-alunite|hydrous_sulphates|sulphates|
-F1CC08B.csv loaded
-bassanite|hydrous_sulphates|sulphates|
-GDS145.csv loaded
-gypsum|hydrous_sulphates|sulphates|
-LASF41A.csv loaded
-jarosite|hydrous_sulphates|sulphates|
-LASF21A.csv loaded
-kieserite|hydrous_sulphates|sulphates|
-F1CC15.csv loaded
-mg-sulphate|hydrous_sulphates|sulphates|
-799F366.csv loaded

Phyllosilicates

-chlorite|chlorites|phyllosilicates|
-LACL14.csv loaded
-kaolinite|kaolinites|phyllosilicates|
-LAKA04.csv loaded
-illite|micas|phyllosilicates|
-LAIL02.csv loaded
-margarite|micas|phyllosilicates|
-GDS106.csv loaded
-serpentine|serpentine|phyllosilicates|
-LALZ01.csv loaded
-montmorillonite|smectites|phyllosilicates|
-LAM002.csv loaded
-nontronite|smectites|phyllosilicates|
-NCJB26.csv loaded
-saponite|smectites|phyllosilicates|
-LASA58.csv loaded
-talc|talcs|phyllosilicates|
-GDS23.csv loaded

Carbonates

-calcite|calcites|carbonates|
-CAGR01.csv loaded

```
-magnesite|calcites|carbonates|  
-F1CC06B.csv loaded
```

Hydrated Silicates & Halides

```
-halite|chlorides|halides|  
-HS433.csv loaded  
-prehnite|prehnites|inosilicates|  
-LAZE03.csv loaded  
-siliceous-sinter|hydrated-silicas|mineraloids|  
-bkr1jb329c.csv loaded  
-epidote|epidotes|sorosilicates|  
-GDS26.csv loaded  
-analcime|zeolites|tectosilicates|  
-GDS1.csv loaded
```

Loading 28 of entries complete.

Exporting the Material Collection to CSV and Pickle formats...

We can use the `material_collection` plotting routines to visualise the library.

```
[8]: ax = matcol.plot_profiles(stacked=True, scope='categories', groupby='Species')
```

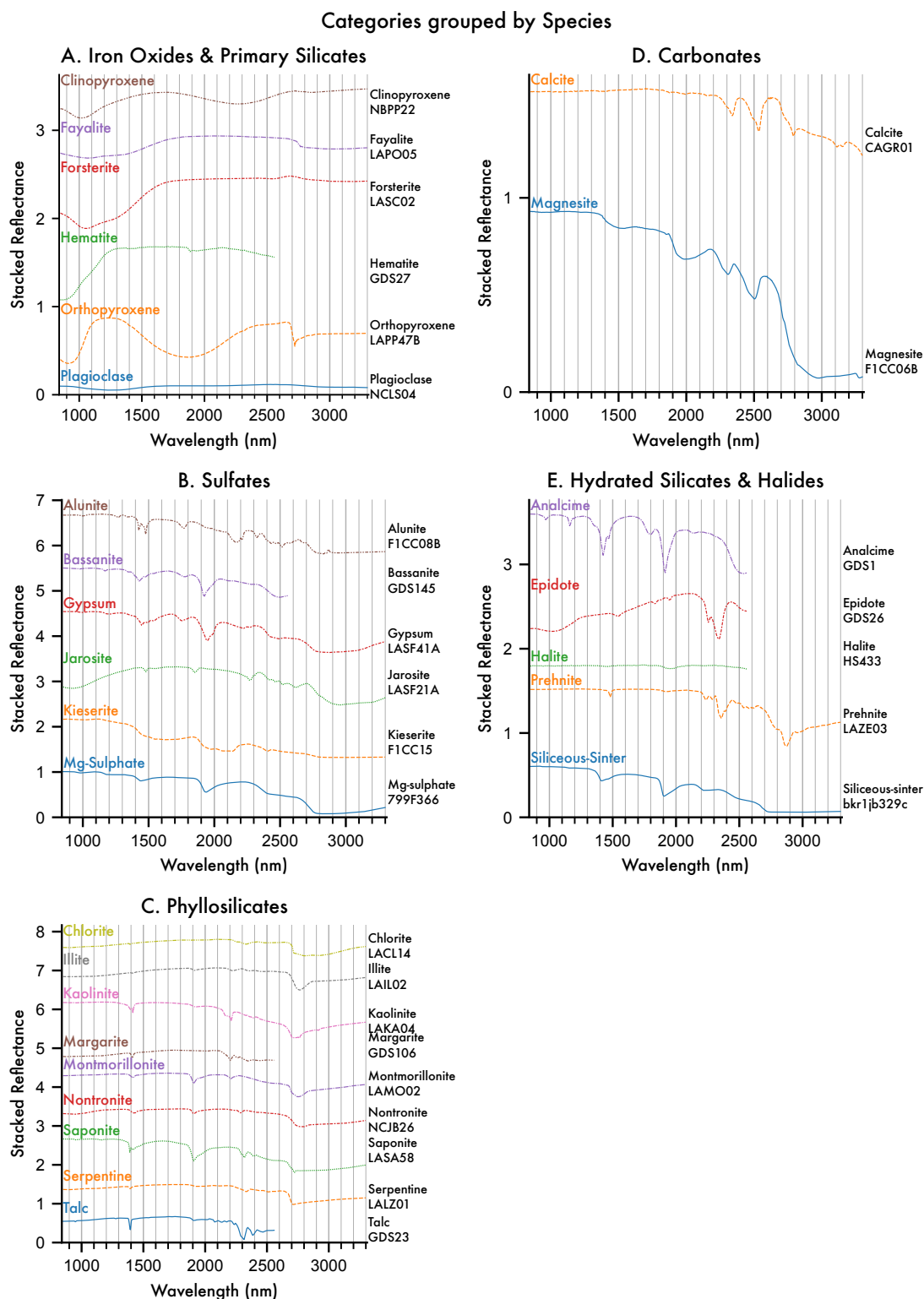


Figure 2 High resolution MICA Files spectral library, arranged by category and grouped by mineral

species.

We can see that not all of the entries cover the complete 0.9 - 3.1 μm range of *Enfys*, so in future work we should look to replace these MICA files laboratory type spectra with spectra that all span the entire range.

We can also produce a waterfall visualisation, an alternative plotting method that presents the information in a condensed format, that is useful for highlighting where absorption features lie.

```
[44]: axes = matcol.plot_profiles(waterfall=True, scope='all', groupby='Category')
```

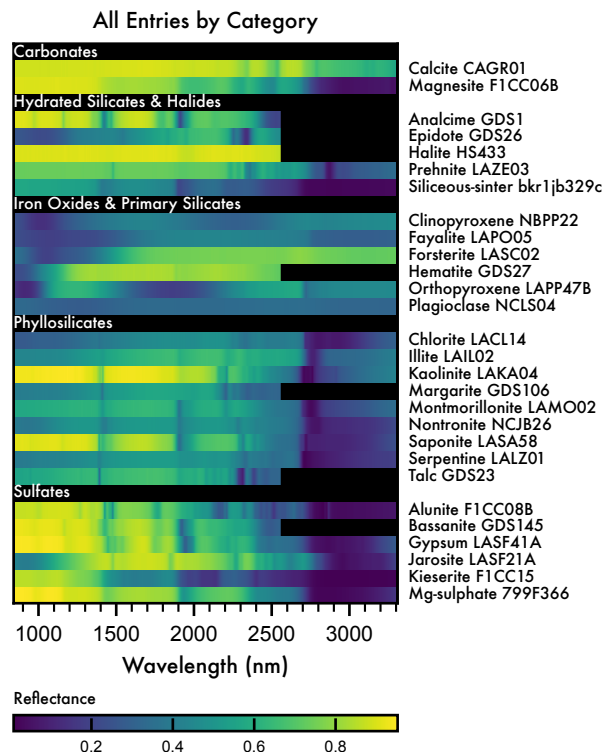


Figure 3 Waterfall plot of the high resolution MICA Files spectral library, arranged by category and grouped by mineral species.

A more useful version of this plot format is to show band locations after continuum removal. We can use the `SpectralLibraryAnalyser.remove_continuum()` function to produce a duplicate of the `MaterialCollection` but with the data continuum-removed.

```
[10]: from sptk.spectral_library_analyser import SpectralLibraryAnalyser as sla

matcol_sla = sla(matcol)
matcol_cr = matcol_sla.remove_continuum()
```

We can plot the profiles to illustrate the continuum removal. Here we use the `scope = 'all'` version of the profile plot.

```
[11]: axes = matcol_cr.plot_profiles(stacked=True, scope='all', groupby='Category')
```

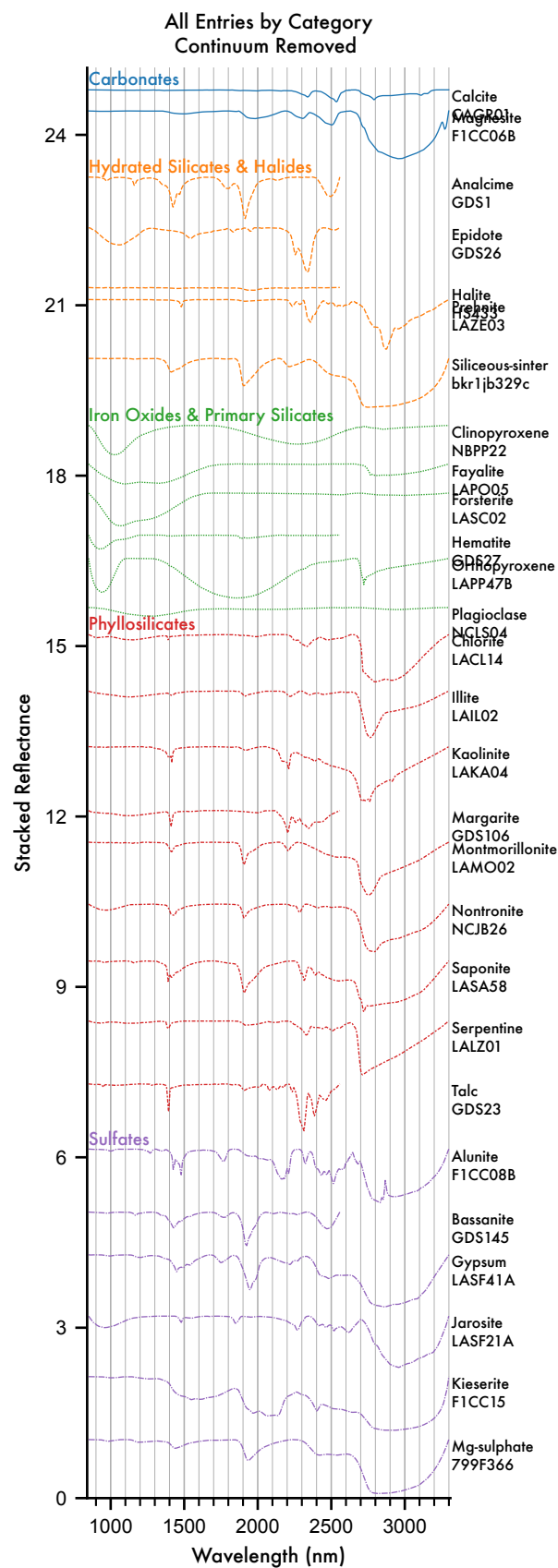


Figure 4 Continuum-removed high resolution MICA Files spectral library, grouped by Category.

Now we plot the waterfall representation:

```
[43]: axes = matcol_cr.plot_profiles(waterfall=True, scope='all', groupby='Category')
```

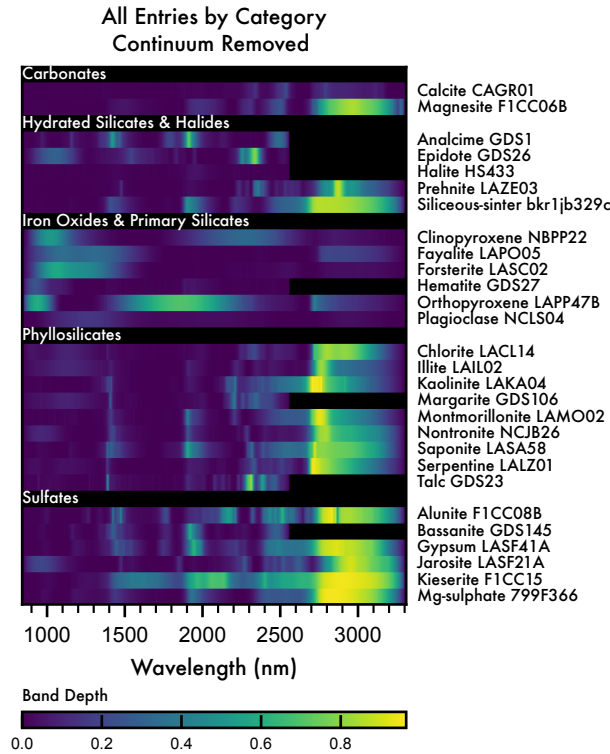


Figure 5 Continuum-removed waterfall plot of high resolution MICA Files spectral library, grouped by Category.

Again, we can see that the waterfall plot represents the band depth information of the `MaterialCollection` in a compact format. Scanning vertically allows for quick identification of common band locations, their depths, widths, and asymmetries.

5 Results

5.1 Simulating the Enfys Spectral Response

To simulate Enfys, rather than providing a file with the expected spectral response of each position of the Linear-Variable Filter (LVF, above), we specify the spectral range and resolution and sampling regime, and generate a set of central-wavelengths and bandwidths. We do this with the `InstrumentBuilder` class of the `sptk.instrument` module.

```
[13]: from sptk.instrument import InstrumentBuilder
```

The expected spectral resolution of Enfys is $\lambda/\Delta\lambda = 100$, and the range is 0.9 - 3.1 μm .

```
[14]: enfys_builder = InstrumentBuilder(  
    instrument_name='enfys',  
    instrument_type='lvf',  
    sampling='nyquist',  
    resolution = 100,  
    spectral_range=[900, 3101],  
    snr_range = [2000, 100])
```

Exporting instrument to ../data/instruments/enfys.csv...

```
[15]: from sptk.instrument import Instrument
```

Now we build the transmission profiles for each LVF position, according to the list of CWLs and FWHMs generated above. These are accessed by reading the .csv file produced by the InstrumentBuilder procedure.

When generating the profiles, we turn the CWL and FWHM into a profile by assuming a distribution function. For Enfys we use the Cauchy distribution, which behaves similarly to a Gaussian, but with wider ‘wings’. It is parameterised by the full-width-at-half-maximum, as the distribution strictly has no finite variance. We have implemented a build_cauchy_filter method for the Instrument class.

```
[16]: enfys = Instrument(  
    name = 'enfys', # filename of the instrument information, held in /  
    ↪software/data/instruments/  
    project_name = matcol.project_name, # the project directory name,   
    ↪hosting outputs in the /spectral_parameter_studies/ directory  
    shape = 'cauchy',  
    load_existing=False, # if True, load existing instrument data from   
    ↪the project directory  
    plot_profiles=False, # if true, plot the transmission profiles   
    ↪during construction  
    export_df=True) # if True, export the instrument transmission   
    ↪profiles to the project directory
```

Building Instrument...

Building new DataFrame for 'enfys'...

Exporting the Instrument to CSV and Pickle formats...

```
[17]: fig ,ax = enfys.plot_instrument_characteristics()
```

Plotting Instrument Transmission...

Plotting Signal-to-Noise Ratio...

Plotting Spectral Resolution...

Plotting FWHM...

Plots exported to ../projects/enfys_mica/instrument

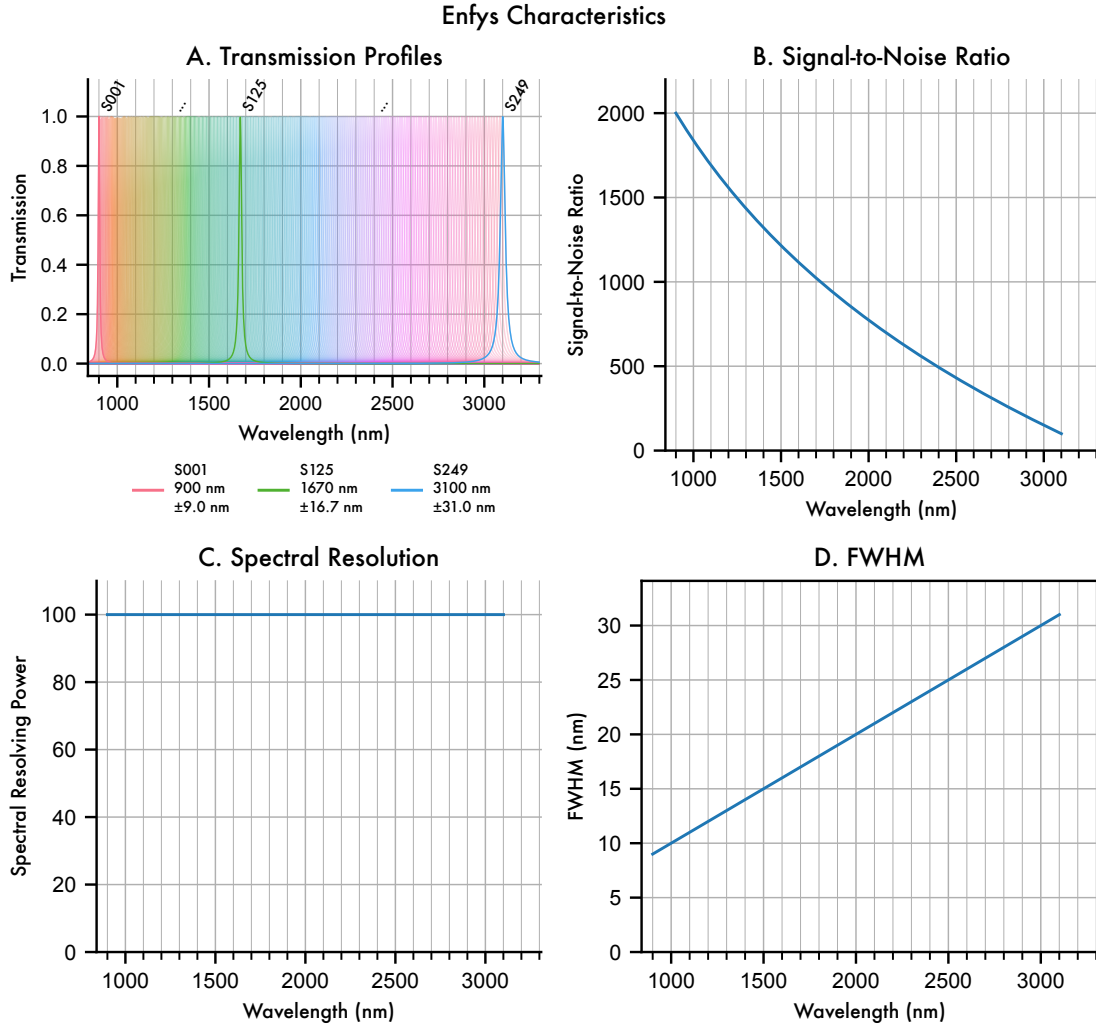


Figure 6 Simulated spectral response and resolution of *Enfys*, based on preliminary expectations of the spectral and radiometric response.

The data shown in figure 3 approximates the expected performance of *Enfys*. Once the instrument is calibrated at the component and assembled levels, this data can be replaced with empirically acquired data.

5.2 Sampling MICA with Enfys

We now make an `Observation` object by sampling the MICA files `MaterialCollection` with the `Enfys Instrument`.

```
[18]: from sptk.observation import Observation
```

```
[19]: obs = Observation(
    material_collection=matcol,
```

```
instrument = enfys,  
load_existing=False,  
plot_profiles=False,  
export_df=True)
```

Building new Observation DataFrame

for 'enfys_mica'...

Sampling the Material Collection with the Instrument...

Sampling complete.

Exporting the Observation Pickle format...

Observation export complete.

We can plot the sampled spectra against the input high-spectra below:

```
[20]: axes = obs.plot_profiles(scope='categories', groupby='Species', stacked=True,   
    ↪ hires_under=True)
```

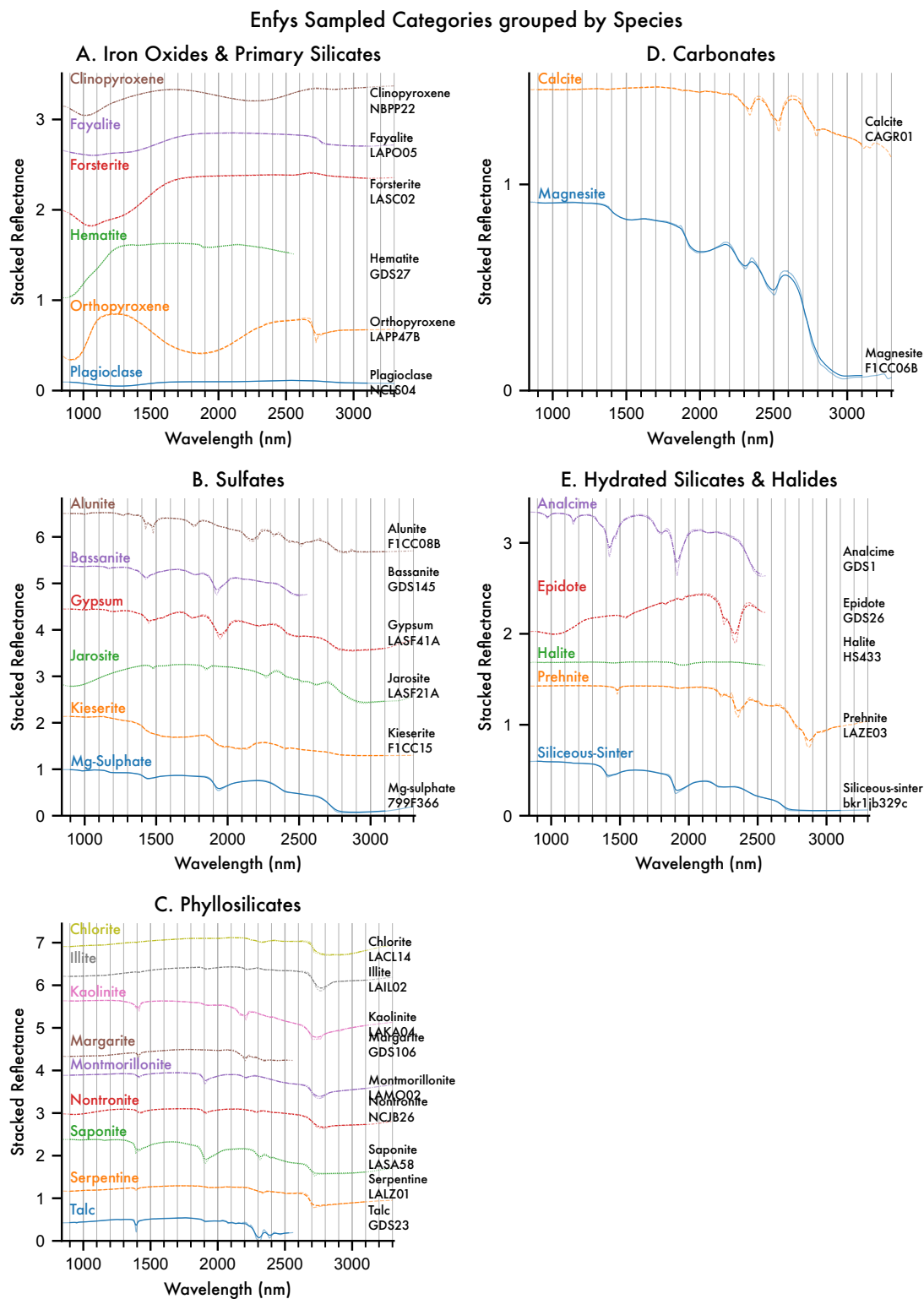


Figure 7 The MICA files as sampled by *Enfys*, with the original high-resolution spectra plotted

underneath each.

We can now see where there are deviations of the *Enfys* sampled spectra from the high-resolution source spectra.

For example, illite, plotted below for clarity.

```
[21]: axes = obs.plot_profiles(scope='illite', groupby='Sample ID', stacked=False,
    ↪ hires_under=True)
```

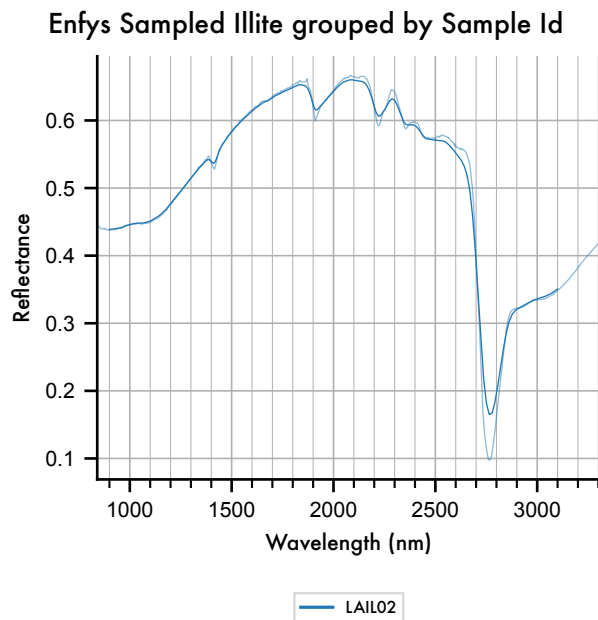


Figure 8 Deviation of the *Enfys* sampled spectrum from the high-resolution input spectrum for Illite.

We note that there is an effect by which peaks and troughs in the spectra are reduced. This is due to the long tails of the Cauchy filter profiles. Despite the reduction of absolute band depth, the features of Illite (fig. 8) are still resolvable.

We can produce a continuum-removed waterfall plot to check that the major bands are resolved.

```
[46]: from sptk.spectral_library_analyser import SpectralLibraryAnalyser as sla
    obs_cr = sla(obs).remove_continuum()
```

```
[47]: axes = obs_cr.plot_profiles(waterfall=True, scope='all', groupby='Category')
```

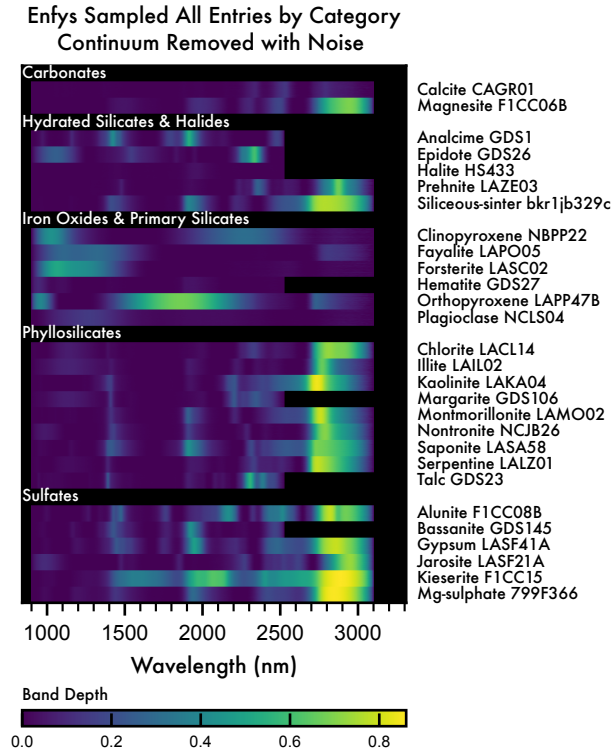


Figure 9 Continuum-Removed waterfall plot of the MICA files as sampled by *Enfys*.

Comparing Figure 9 to Figure 5, we can see that the weaker sub-2 μm bands are less well resolved.

5.3 Adding Synthetic Noise

As discussed in the method, we apply a simple additive Gaussian noise component, by redrawing $n = 100$ additional measurements. We can plot the full range of the resultant samples, as shown again for the example of Illite below.

```
[48]: noise_obs = obs.add_noise(100, seed=0)

[49]: axes = obs.plot_profiles(scope='illite', groupby='Sample ID', stacked=False,
    ↪ hires_under=True, ci=False)
```

Enfys Sampled Illite grouped by Sample Id with Noise

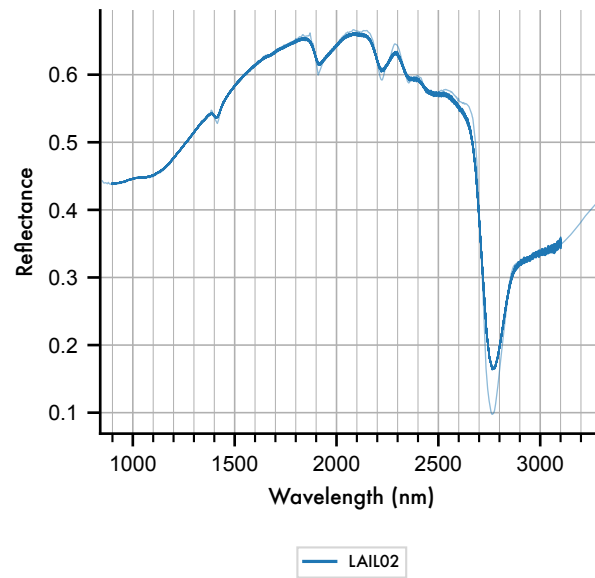


Figure 10 Noisy resampled example spectrum for Illite, showing 100 repeat samples according to the spectral signal-to-noise ratio illustrated in figure 3.B.

We can also plot the mean observed line, with shading indicating the standard deviation of the noisy data, shown below in figure 7, where the 'ci' keyword has been applied ('ci': confidence-interval).

```
[50]: axes = obs.plot_profiles(scope='illite', groupby='Sample ID', stacked=False,
    ↳ hires_under=True, ci=True)
```

Enfys Sampled Illite grouped by Sample Id with Noise

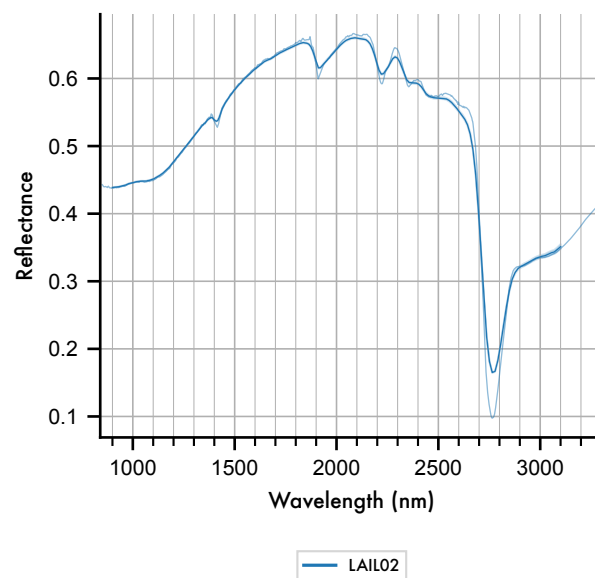


Figure 11 Mean of Noisy resampled example spectrum for Illite, showing mean and standard deviation of 100 repeat samples according to the spectral signal-to-noise ratio illustrated in figure 3.B.

Here we find that the standard deviation is almost negligible at the scale of the figure.

Plotting out all of the groups, we again see that the noise, at this scale, is indeed negligible for most entries. Here we opt to plot the complete range of noisy samples, rather than averaging and showing the (very small) standard deviation.

```
[51]: axes = obs.plot_profiles(scope='categories', groupby='Species', stacked=True,
    ↪ hires_under=True, ci=False)
```

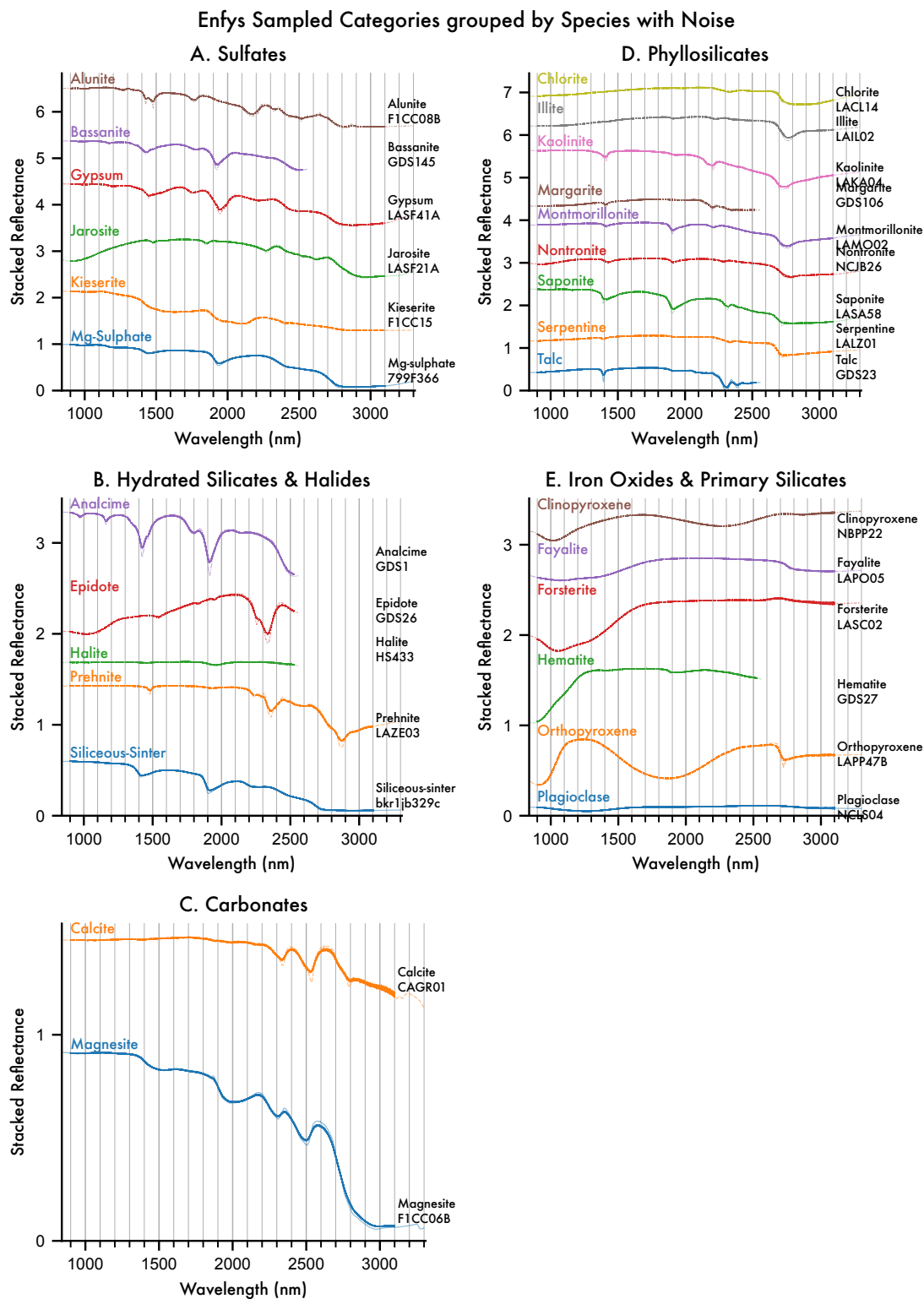


Figure 12 The MICA files as sampled by *Enfys*, with repeat samples drawn according to the ex-

pected spectral Signal-to-Noise Ratio, with the original high-resolution spectra plotted underneath each.

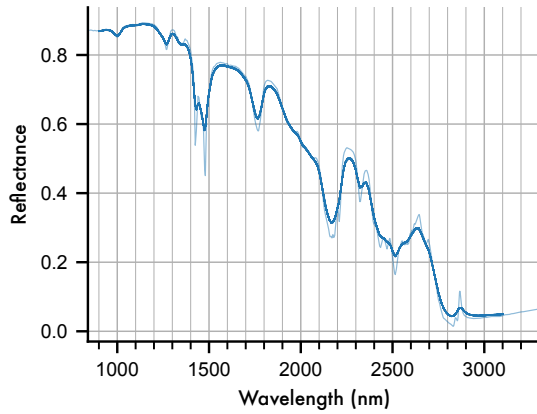
The key conclusion to draw from the above figure is that expected noise for *Enfys* does not significantly intrude on the diagnostic features of these type specimen of minerals identified on the Mars surface.

We can illustrate this in greater deal by plotting each entry on a separate figure, with the y-axis clipped to the spectral range of entry. Note that for sample with a small spectral range (i.e. samples that are spectrally flat), this stretching will appear to amplify noise, so the y-axis range must be considered when assessing the noise.

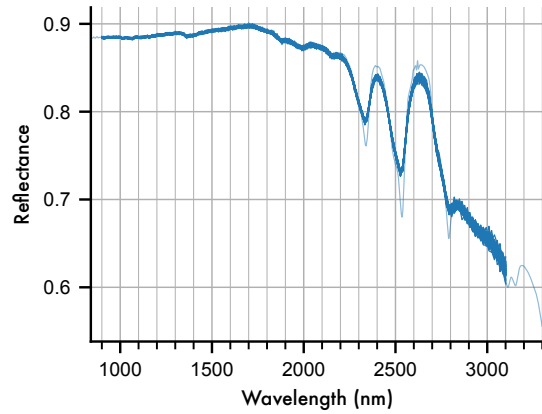
```
[59]: axes = obs.plot_profiles(scope='samples', groupby='Species', stacked=False,
    ↪ hires_under=True, ci=False)
```

Enfys Sampled Samples grouped by Species with Noise (1/5)

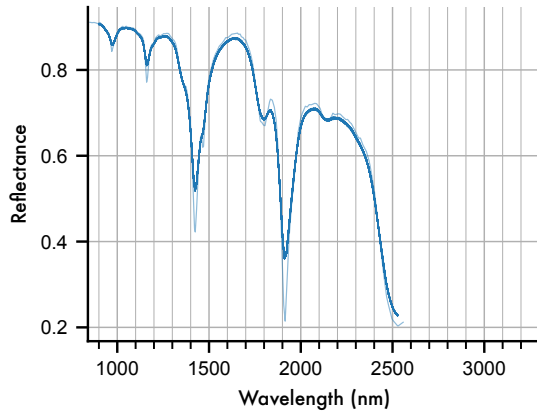
A. F1Cc08B



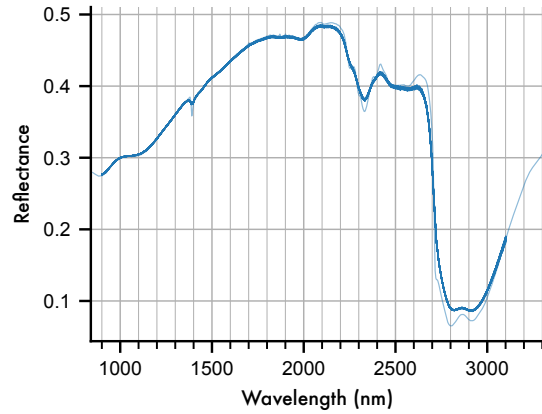
D. Cagr01



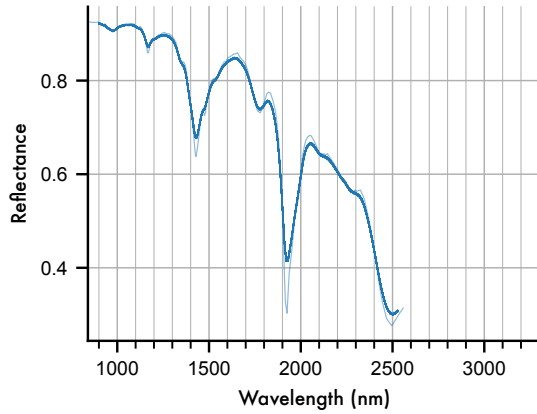
B. Gds1



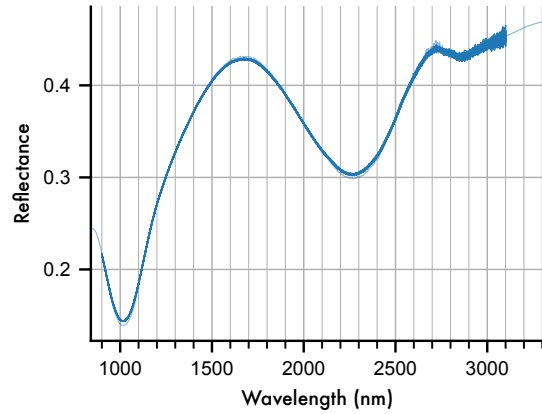
E. Lacl14



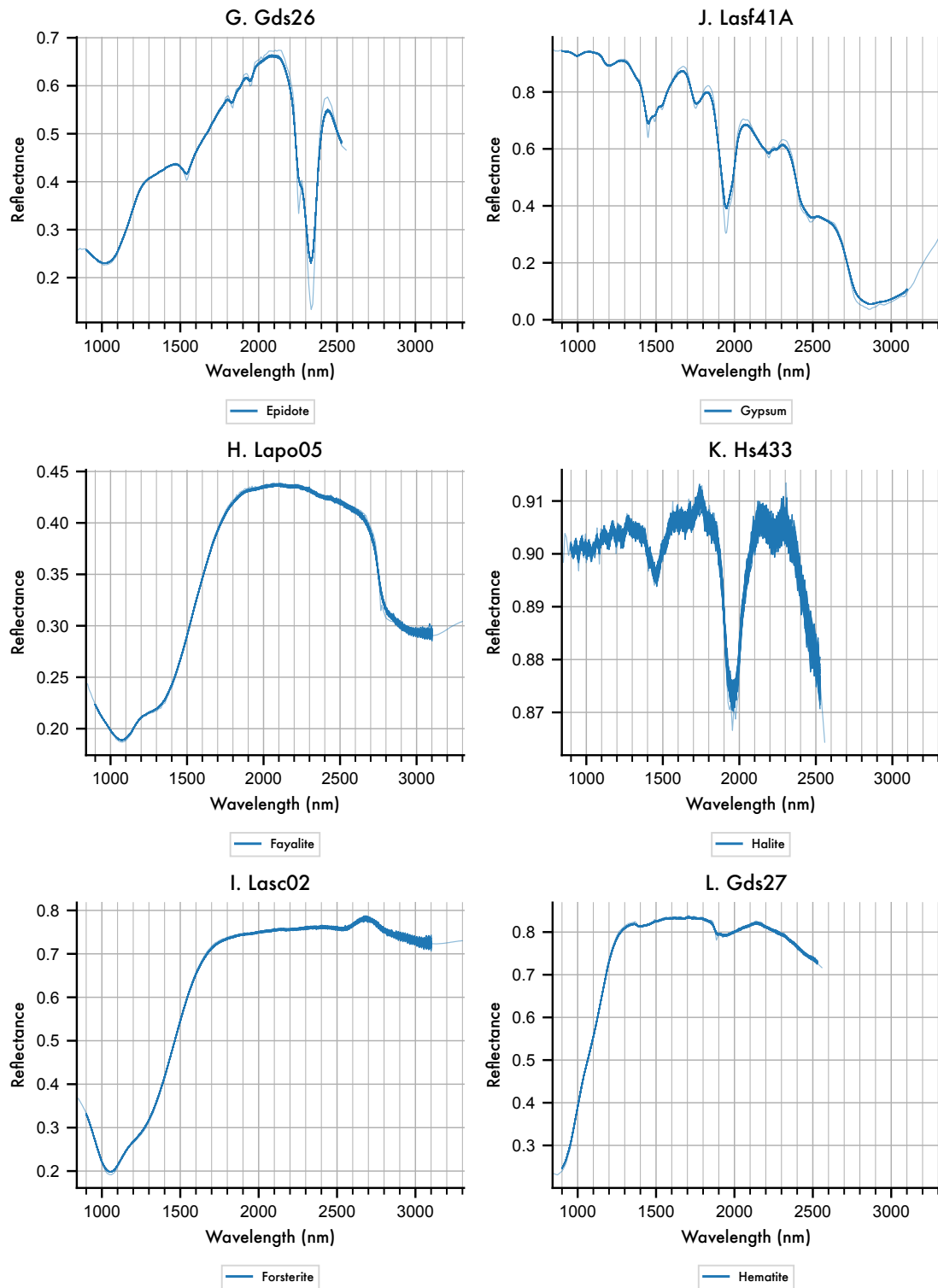
C. Gds145



F. Nbpp22

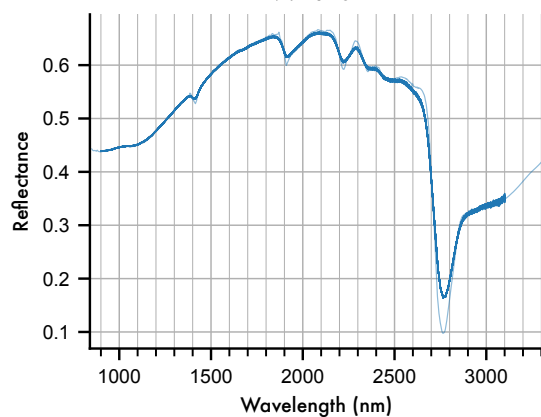


Enfys Sampled Samples grouped by Species with Noise (2/5)

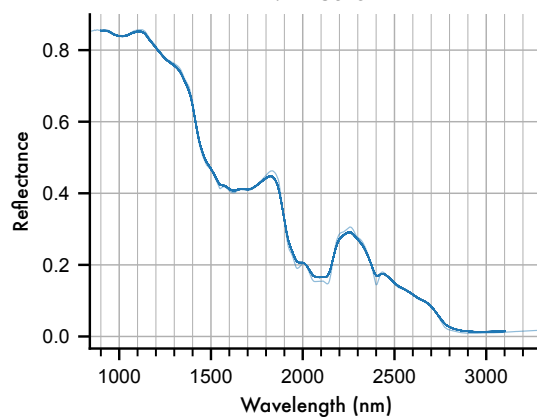


Enfys Sampled Samples grouped by Species with Noise (3/5)

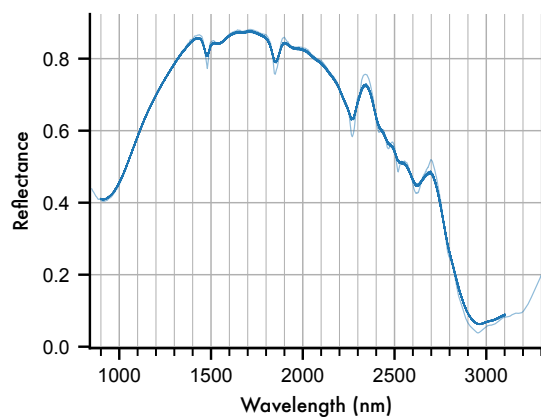
M. Lai02



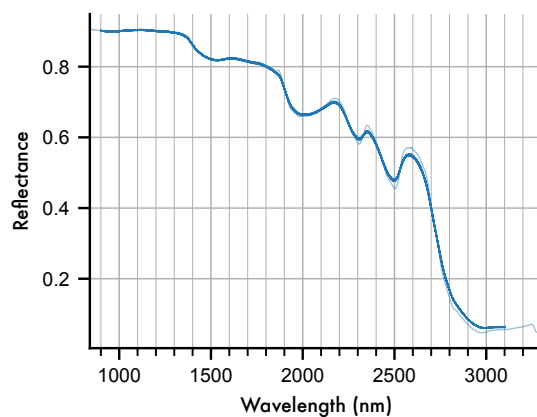
P. F1Cc15



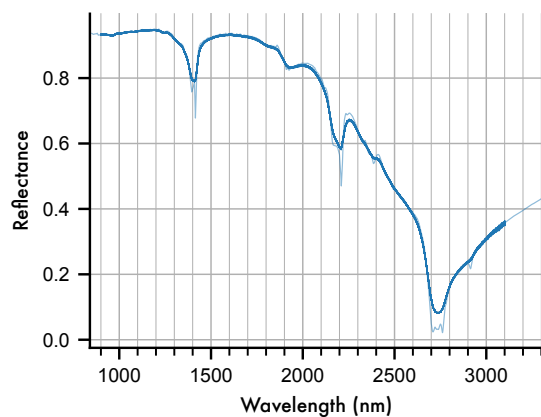
N. Lasf21A



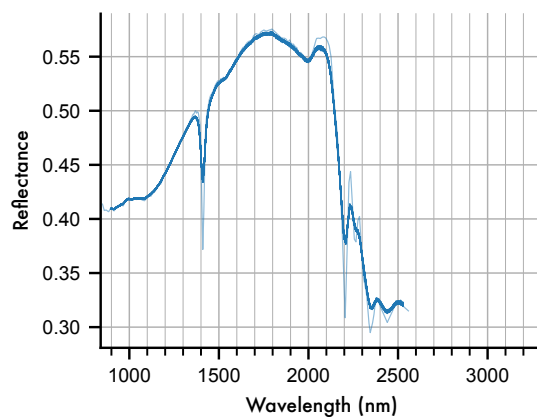
Q. F1Cc06B



O. Laka04

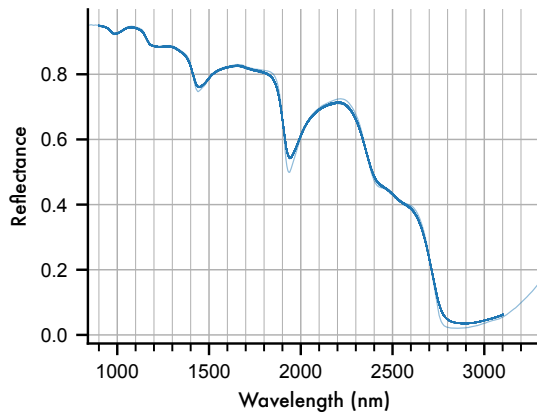


R. Gds106



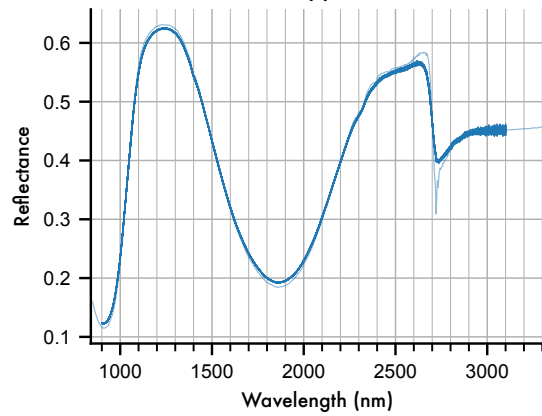
Enfys Sampled Samples grouped by Species with Noise (4/5)

S. 799F366



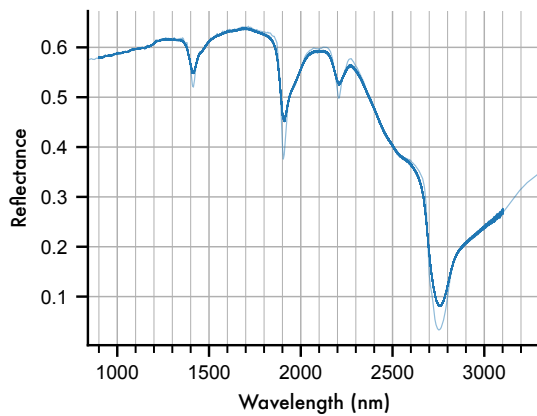
Mg-Sulphate

V. Lapp47B



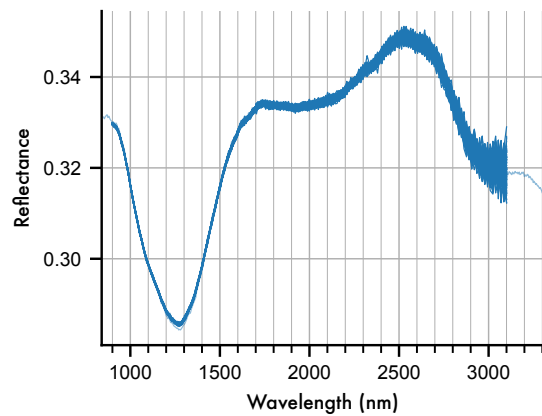
Orthopyroxene

T. Lamo02



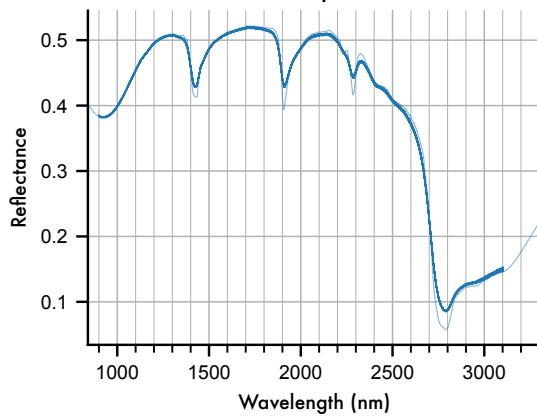
Montmorillonite

W. Ncls04



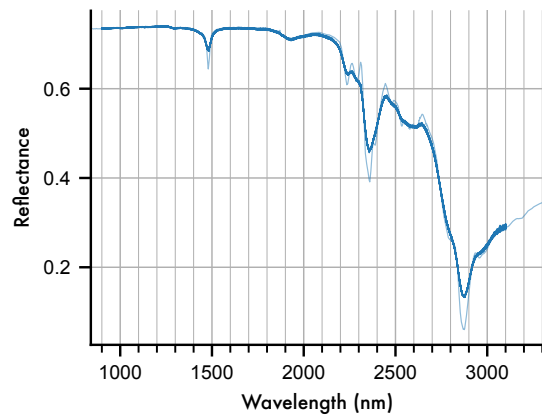
Plagioclase

U. Ncjb26



Nontronite

X. Laze03



Prehnite

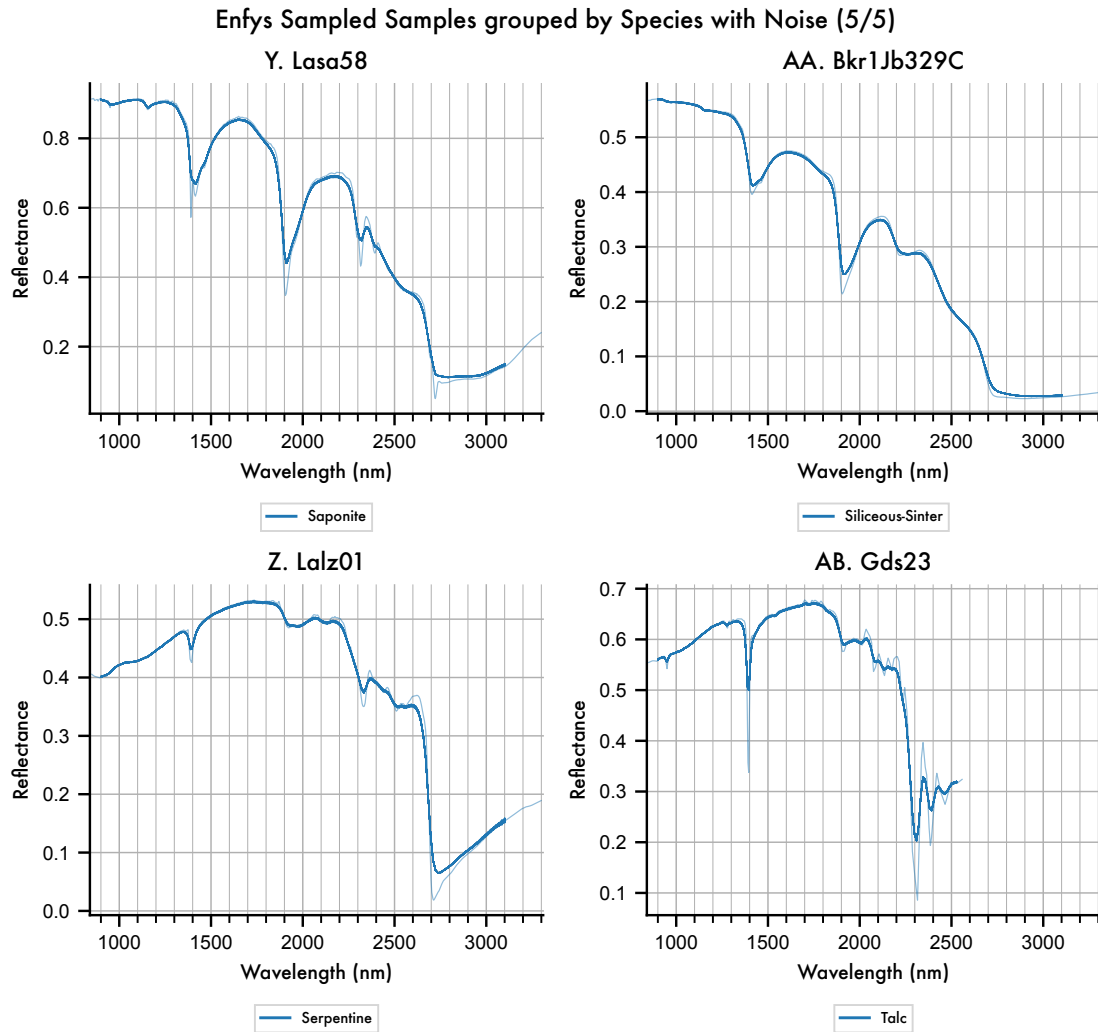


Figure 13 Detailed view of each entry in the spectral library, showing the scale of noise introduced, relative to the features of each spectrum, and the reflectance range.

5.4 Continuum Removal

We can apply continuum removal, even to the noisy data.

```
[53]: from sptk.spectral_library_analyser import SpectralLibraryAnalyser as sla

      cr_tool = sla(obs)
      cr_obs = cr_tool.remove_continuum()

[54]: axes = cr_obs.plot_profiles(scope='categories', groupby='Species',
      ↪stacked=True, hires_under=False, ci=False)
```


Enfys Sampled Categories grouped by Species
Continuum Removed with Noise

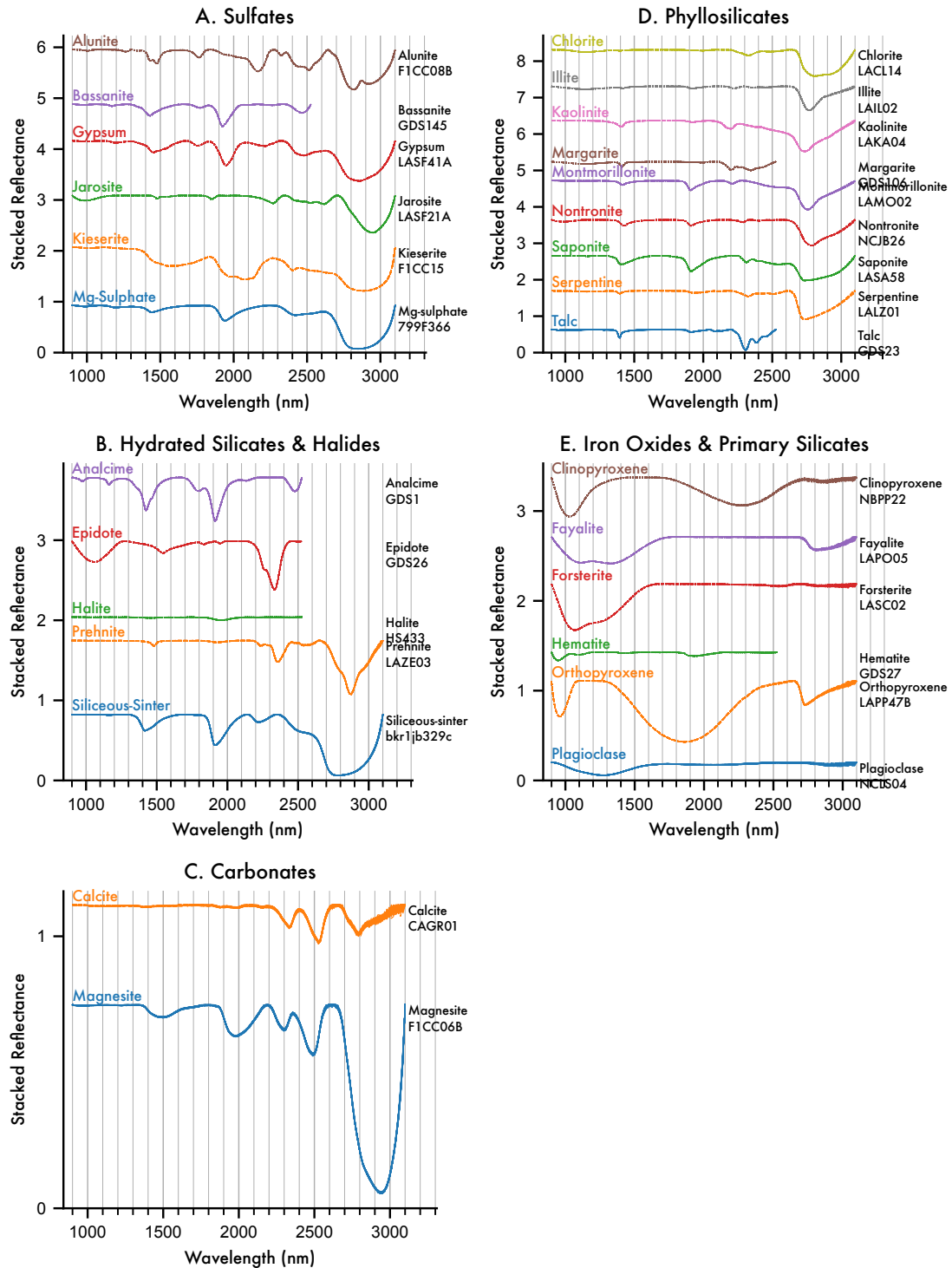


Figure 14 Continuum-Removed entries of the MICA Files, as sampled by *Enfys*, including syn-

thetic noise.

5.5 Waterfall Visualisation

```
[56]: axes = obs.plot_profiles(scope='all', groupby='Category', stacked=False,
    ↪waterfall=True, ci=False)
```

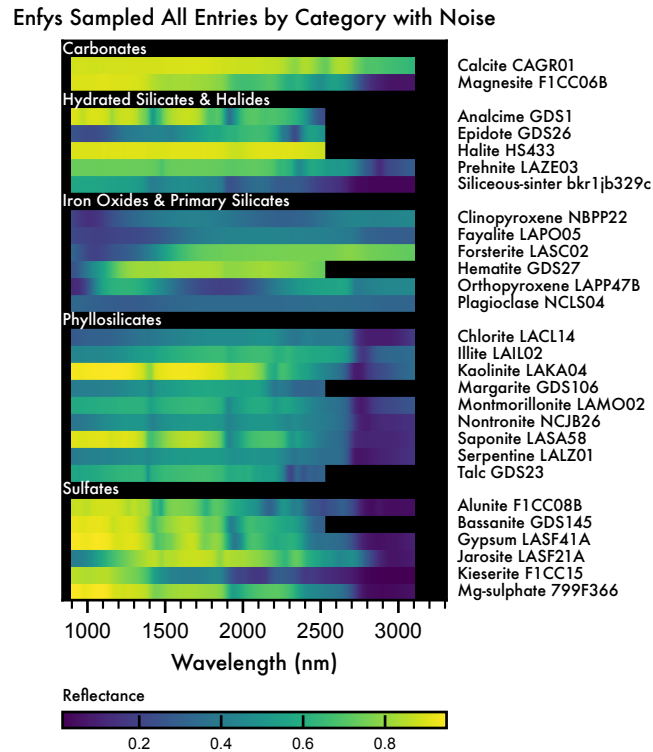


Figure 15 Waterfall diagram of the entries of the MICA Files, as sampled by *Enfys*, including synthetic noise. Each entry bar is composed of n repeat noisy samples, such that each bar is composed of n rows, illustrating the noise.

```
[57]: axes = cr_obs.plot_profiles(scope='all', groupby='Category', stacked=False,
    ↪waterfall=True, ci=False)
```

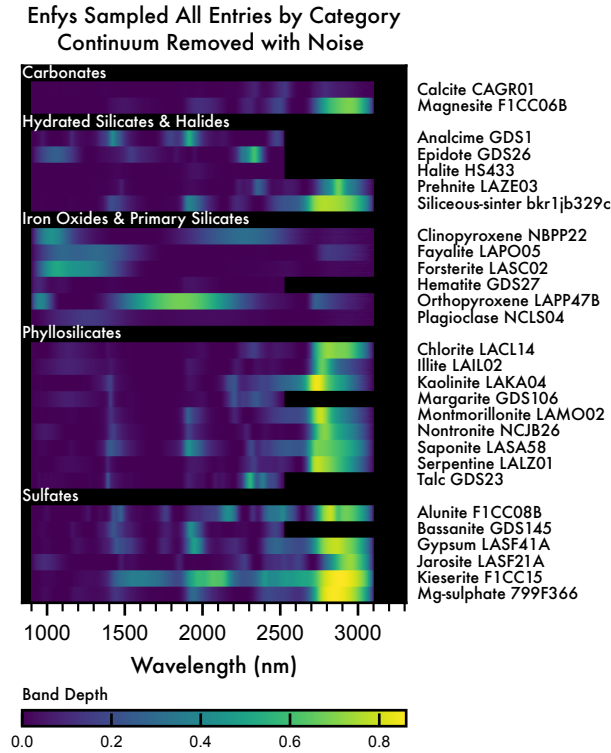
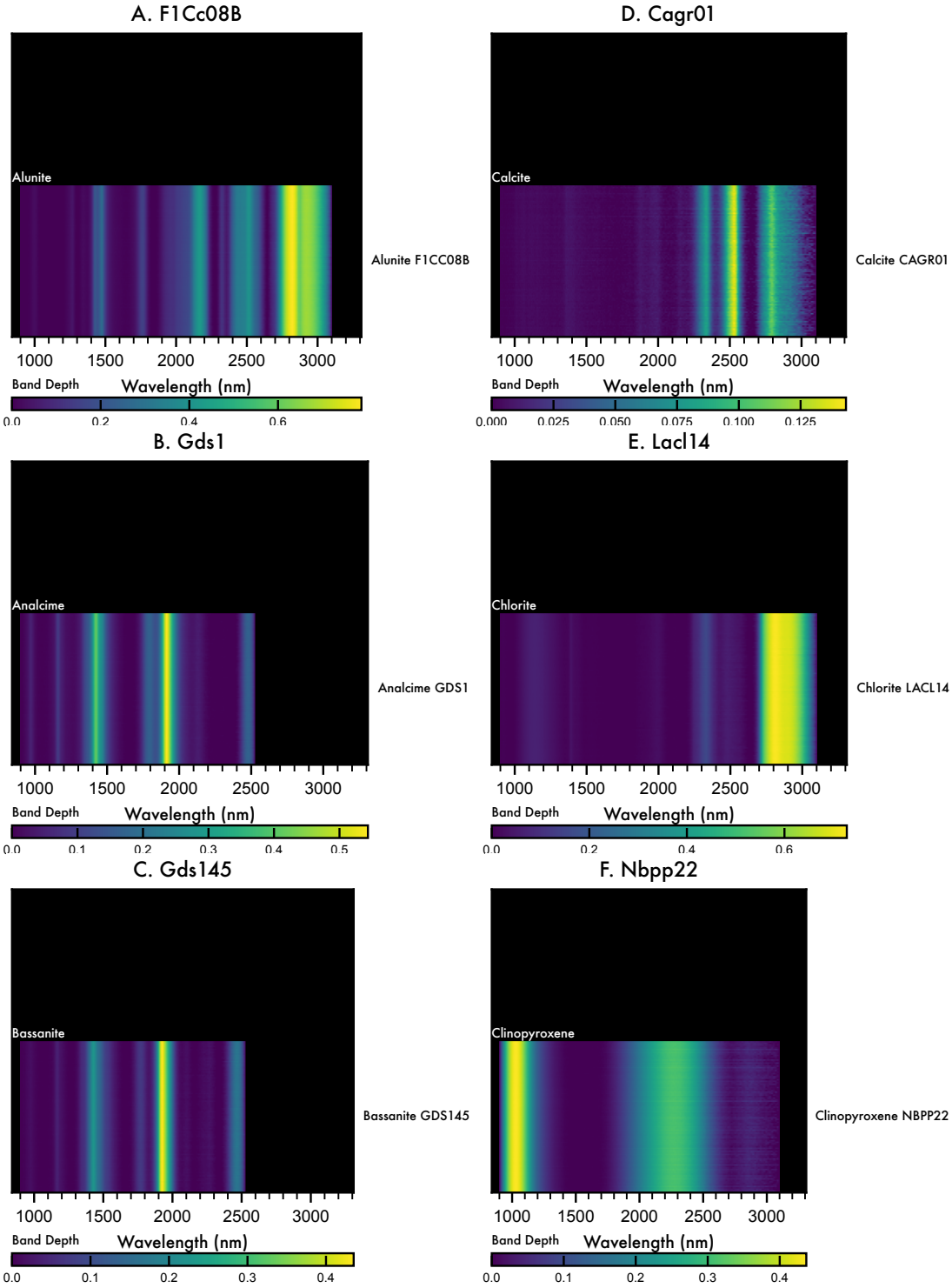


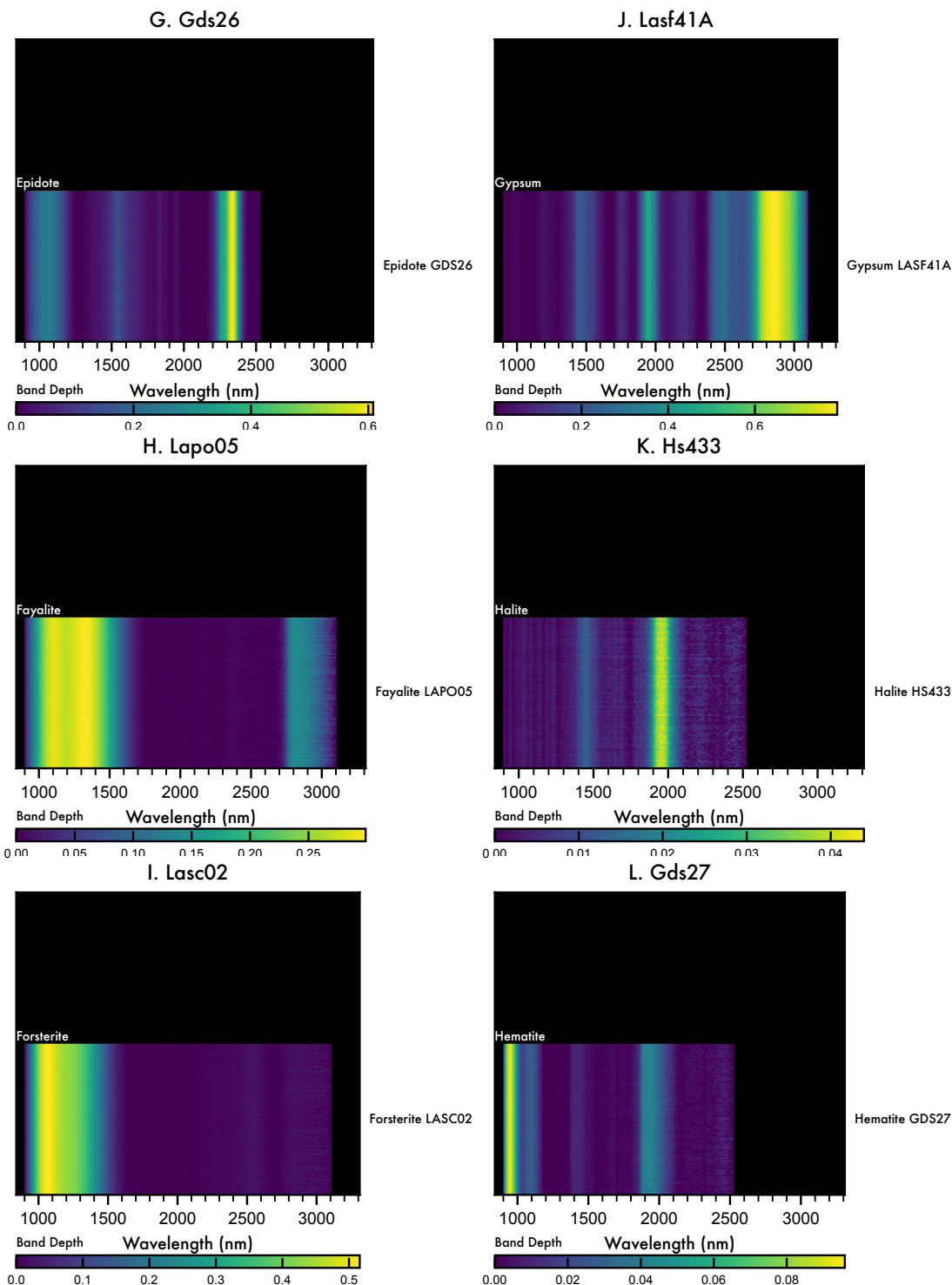
Figure 16 Continuum-Removed waterfall diagram of the entries of the MICA Files, as sampled by *Enfys*, including synthetic noise. Each entry bar is composed of n repeat noisy samples, such that each bar is composed of n rows, illustrating the noise.

```
[58]: axes = cr_obs.plot_profiles(scope='samples', groupby='Species', stacked=False,
    ↪waterfall=True, ci=False)
```

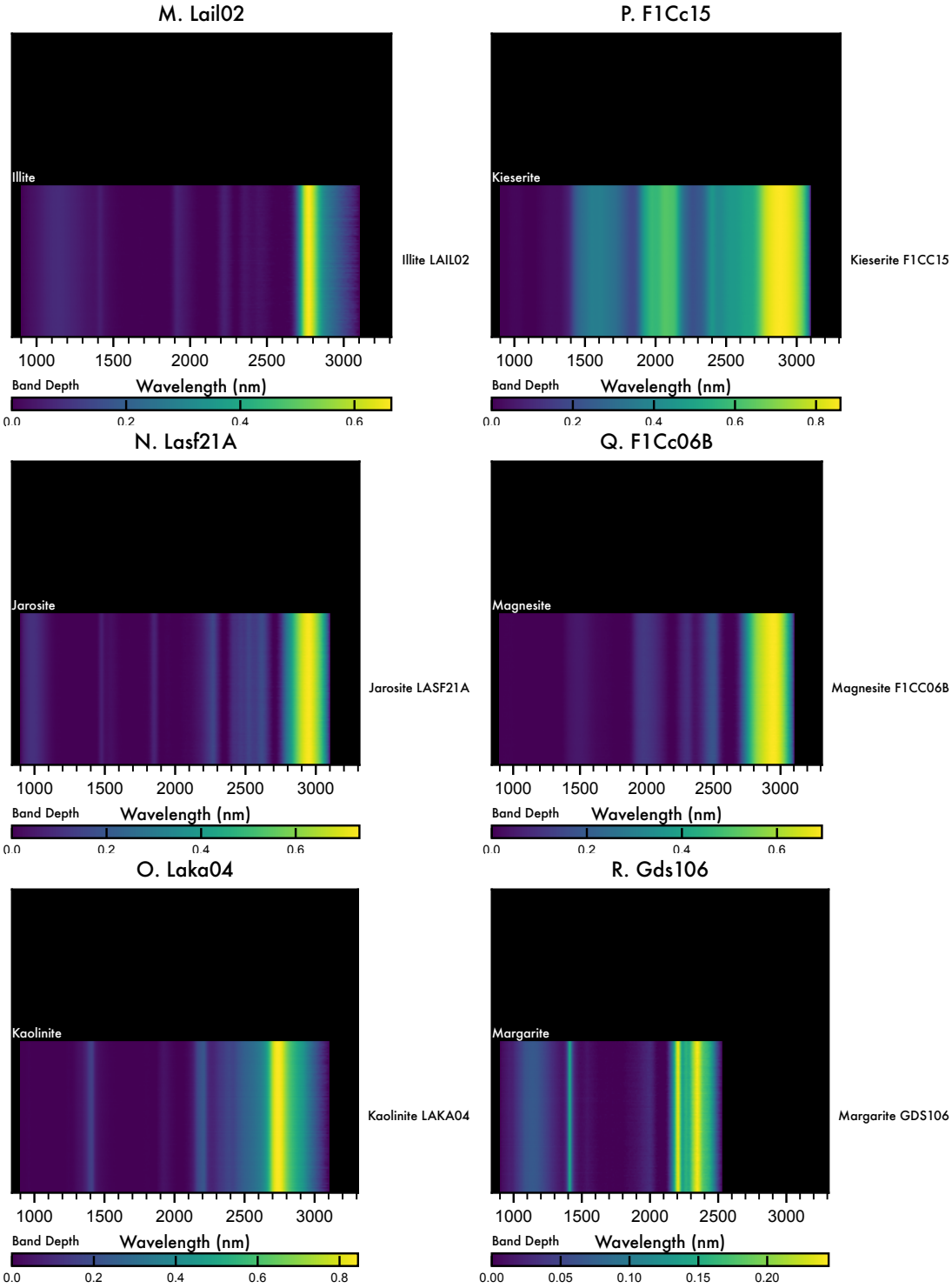
Enfys Sampled Samples grouped by Species
Continuum Removed with Noise (1/5)



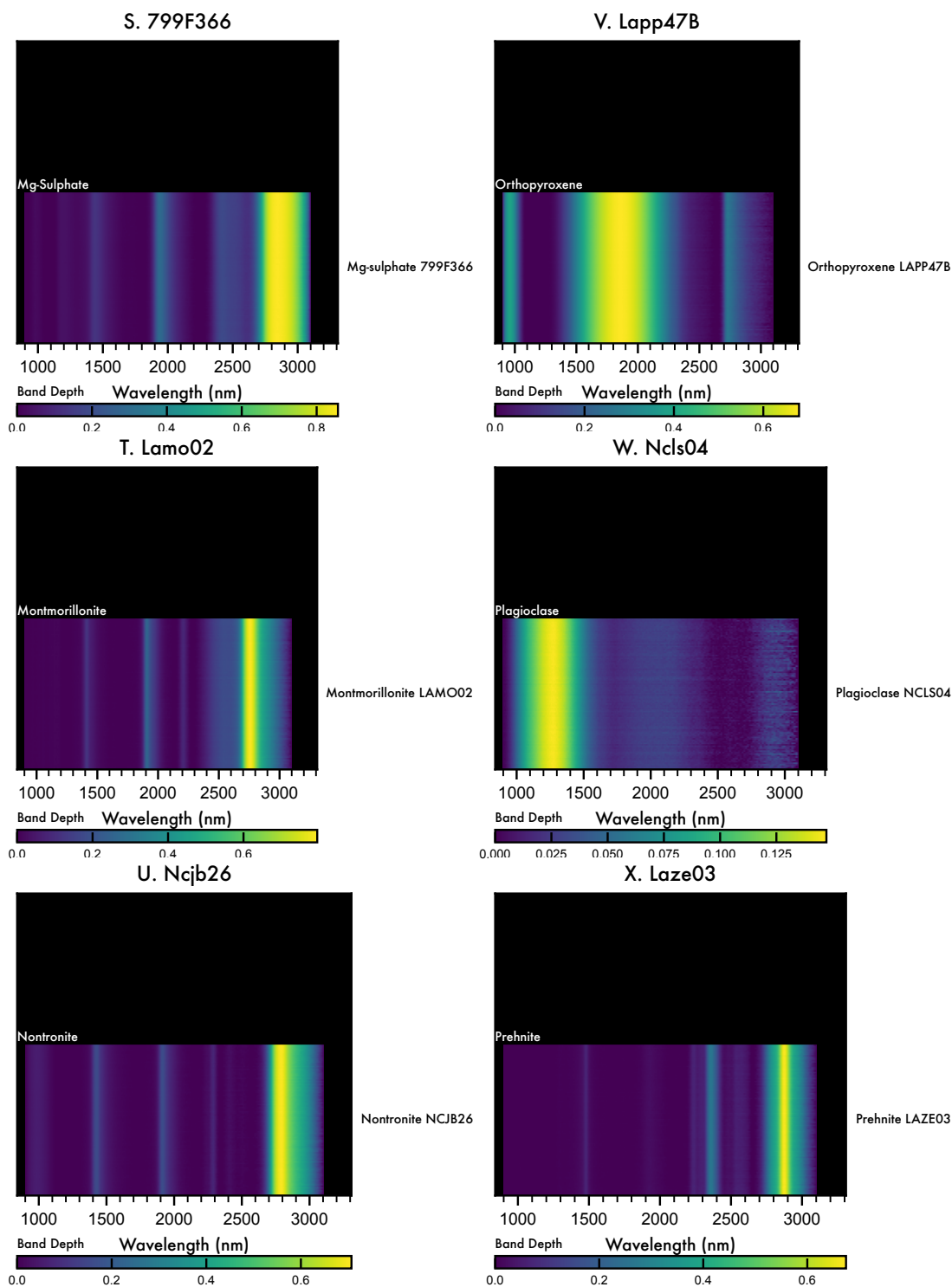
Enfys Sampled Samples grouped by Species
Continuum Removed with Noise (2/5)



Enfys Sampled Samples grouped by Species
Continuum Removed with Noise (3/5)



Enfys Sampled Samples grouped by Species
Continuum Removed with Noise (4/5)



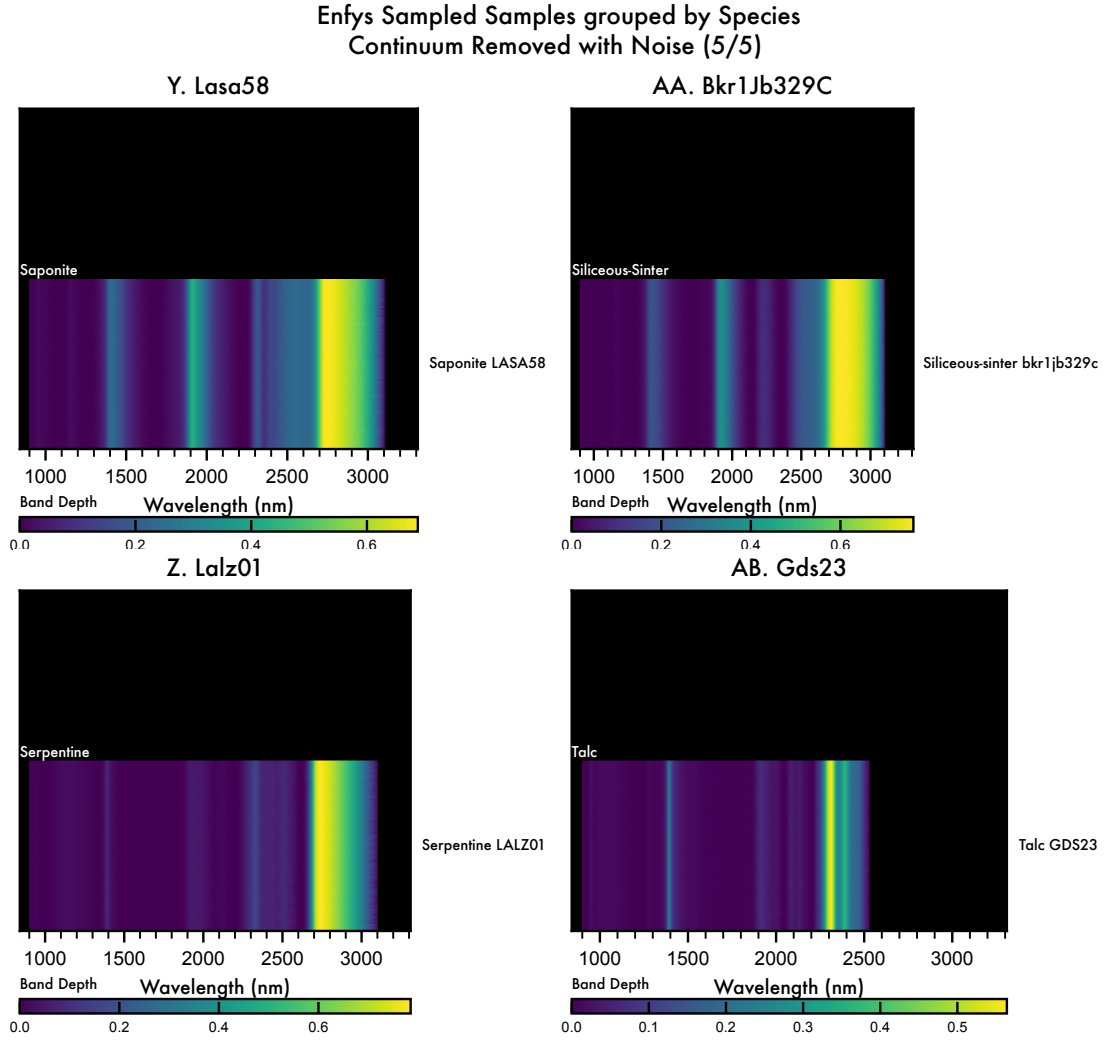


Figure 17 Detailed waterfall view of each entry in the spectral library, showing the scale of noise introduced, relative to the features of each spectrum, and the reflectance range. Each entry bar is composed of n repeat noisy samples, such that each bar is composed of n rows, illustrating the noise.

5.6 Spectral Parameter Evaluations

TODO: get a list of spectral parameter IDs as used for e.g. CRISM, and evaluate the closest ones for Enfys. Then show the spectral parameters in a colour-grid.

6 Discussion

Notes on what we found above, perhaps with some more analysis.

7 Conclusions

Concluding remarks and summary of findings.

8 References